



Scooterrari Knee Scooter Mechanical Design Report



Figure 1. Scooterrari Knee Scooter Final Assembly.



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Design Overview and Relevant Background:

Problem Statement and Design Goal:

For individuals with foot or leg injuries, traditional crutches often cause upper-body fatigue and instability. The goal of this project was to design a knee scooter that provides a stable, ergonomic, and low-resistance alternative. Our design focuses on a 160 lb capacity target, prioritizing a "quality feel" through rigid construction while remaining lightweight enough for easy transport. The final design aims to minimize mechanical deflection and maximize user stability during both standard operation and "abuse cases," such as navigating curb drops which we are modeling as twice the nominal body weight.

How it Works:

The scooter operates on a three-wheel, dual-axle platform. The user places their injured leg's knee on a padded, height-adjustable seat while using their healthy leg to propel themselves forward. Steering is managed via a handlebar column connected to the front axle. This configuration allows for a natural, upright posture, reducing strain on the back and armpits compared to crutches.

Target Audience and Use Constraints:

- Target Audience: Primarily designed for a 5'10", 160 lbf male, though the adjustable seat allows for a range of heights and similar weight profiles.
- Who Shouldn't Use This: Individuals exceeding the 160 lbf weight limit or those with severe balance impairments. It is not intended for use on soft sand, deep gravel, or steep inclines, as the 8-inch wheels are optimized for flat, paved surfaces like concrete or asphalt.

Design Constraints and Criteria:

To ensure the scooter is both functional and portable, the following specifications were established:

Operational and Construction Challenges:

Table 1. Design Criteria and Target Specifications

Design Criteria	Specification
Weight Capacity	<160 lbs
Deflection Limit	<0.25 in
Width	<12 in
Scooter Weight	<25 lbs
Factor of Safety	>2

Concepts and Ideation:

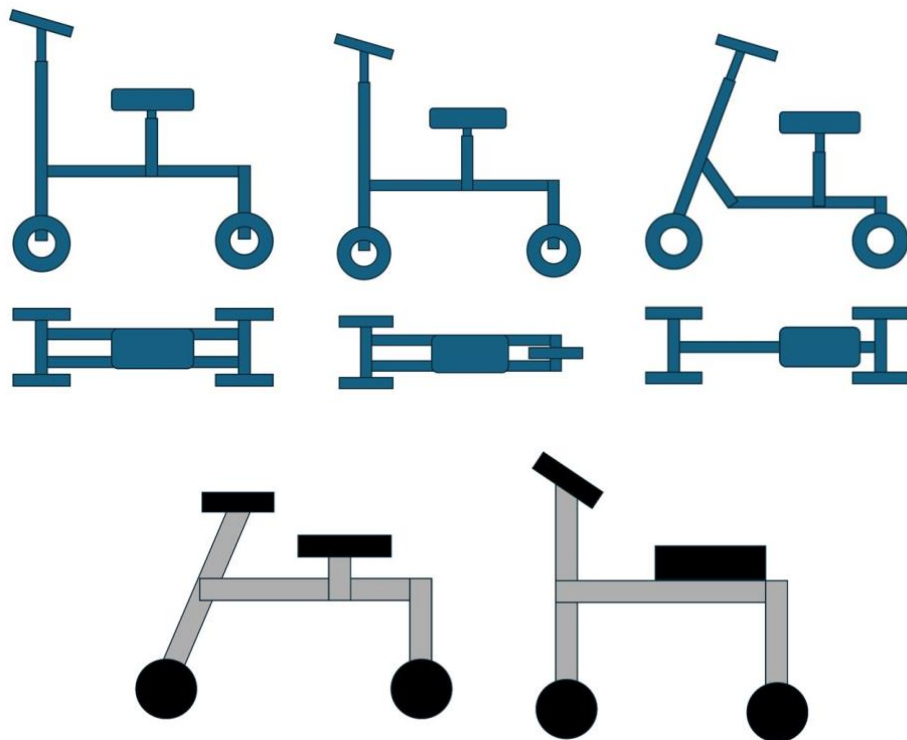


Figure 2. Scooterarri Knee Scooter Concepts.



As shown in figure 2, we came up with 5 core concepts which we then discussed to land on our final design. We discussed an angled steering column to increase the scooters ability to approach bumps and downhill terrain, but decided against it because it would not work well with two wheels in the front. Our final design was the second design with two parallel members, three wheels, a vertical steering column, and a vertically adjustable seat post. We chose the parallel cross members because we wanted to do go with a different design than the example and make an original design. Throughout the design process, we decided to not make an adjustable steering column because that would have decreased the size of the tube for a portion of the column, increasing the deflection. We went with three wheels to decrease the cost and weight of the scooter, but we left two in the front to keep the necessary stability for the user. We went with an adjustable seat post to allow a wider range of users to comfortably ride it.

Construction, Operation, and Safety Challenges and Concerns:

Construction and Material Integrity:

The use of PVC piping as a primary structural element required precise cutting and to ensure flush fits within joints. Any misalignment in the dual-beam platform can lead to uneven weight distribution, potentially compromising the calculated factor of safety. Making sure the horizontal beams were exactly parallel to each other, and that the wheel columns were aligned required careful planning.

Operational Stability and Ergonomics:

FEA modeling indicated a high theoretical displacement under peak bending stress for the steering column. While this is an extreme limit, it suggests that users may experience a "spongy" steering feel during heavy maneuvering.

While the design is optimized for a 160 lb capacity, the "abuse case" (modeling twice the nominal weight for curb drops) presents a significant challenge. We expect the scooter to likely fail under such conditions.

Safety Concerns:

Without proper bearings and tolerance control, the scooter currently has a tendency to want to tip over. Along with this, the scooter is designed for flat, paved surfaces. Because of this, it is highly against our recommendations to take the scooter offroad or on steep inclines.

Drawings and Assembly:

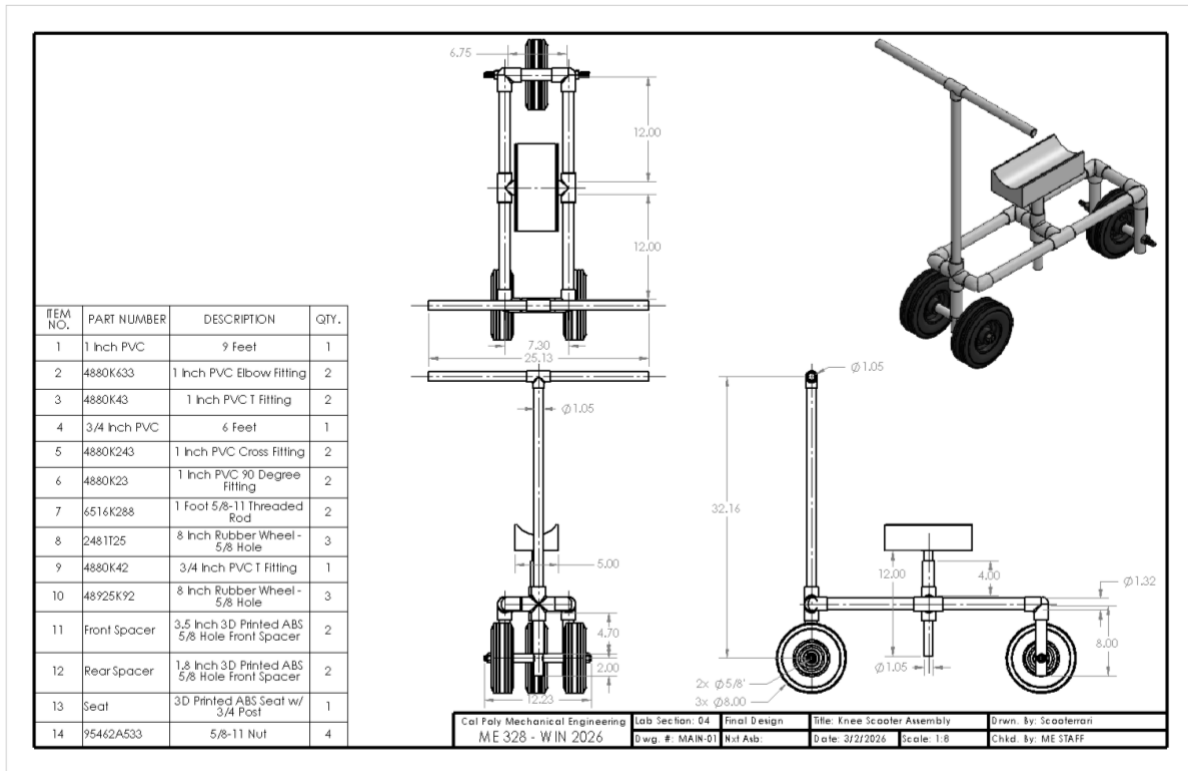


Figure 3. Knee Scooter Assembly Drawing and Bill of Materials.

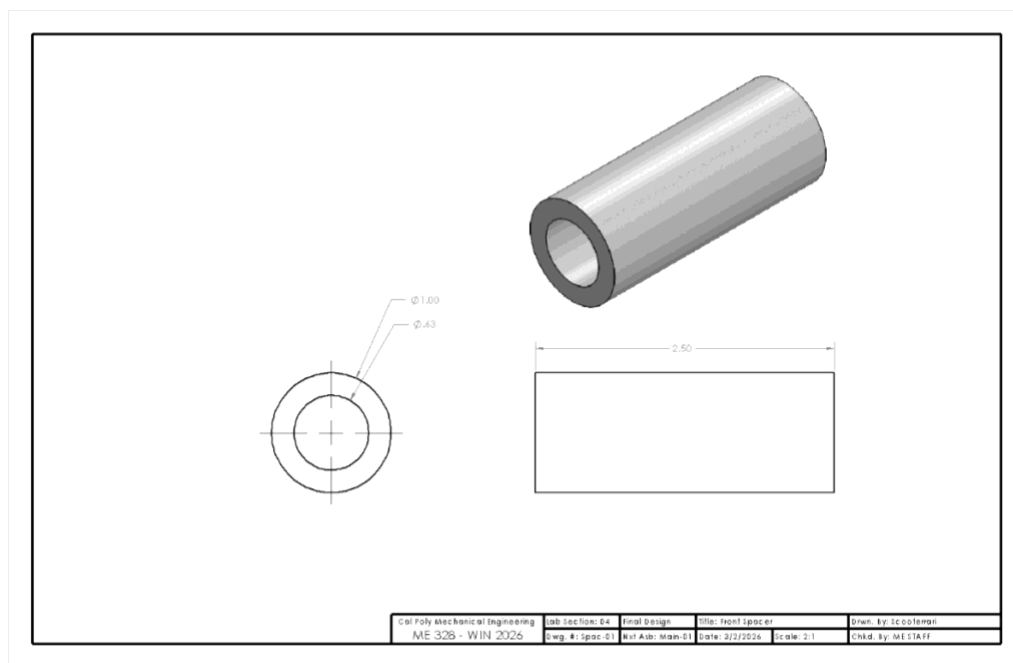


Figure 4. 3D Printed Front Axle Spacer Drawing.

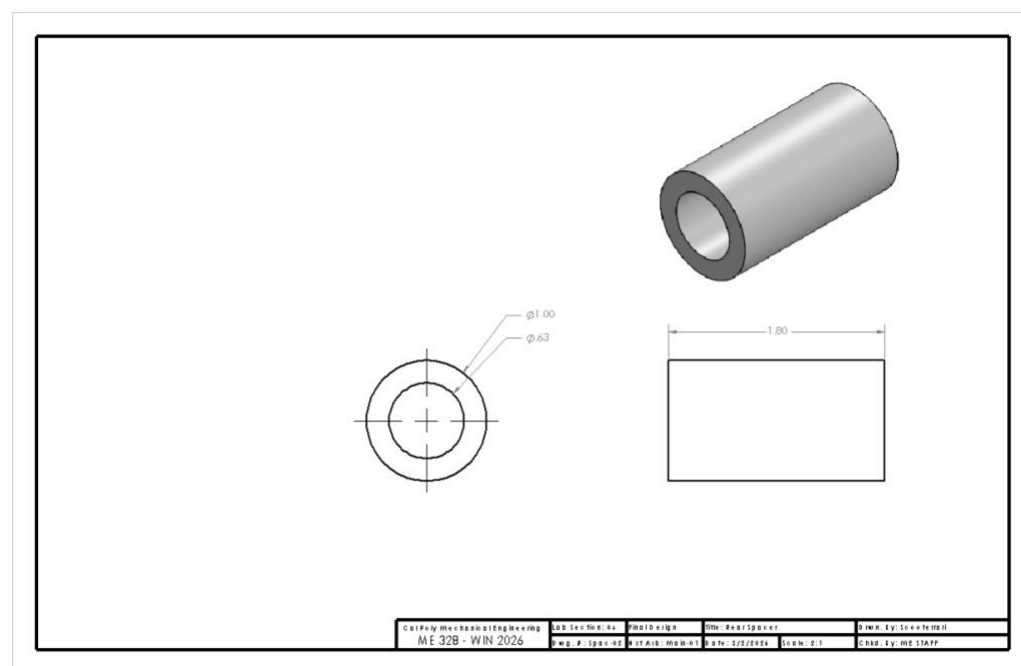


Figure 5. 3D Printed Rear Axle Spacer Drawing.

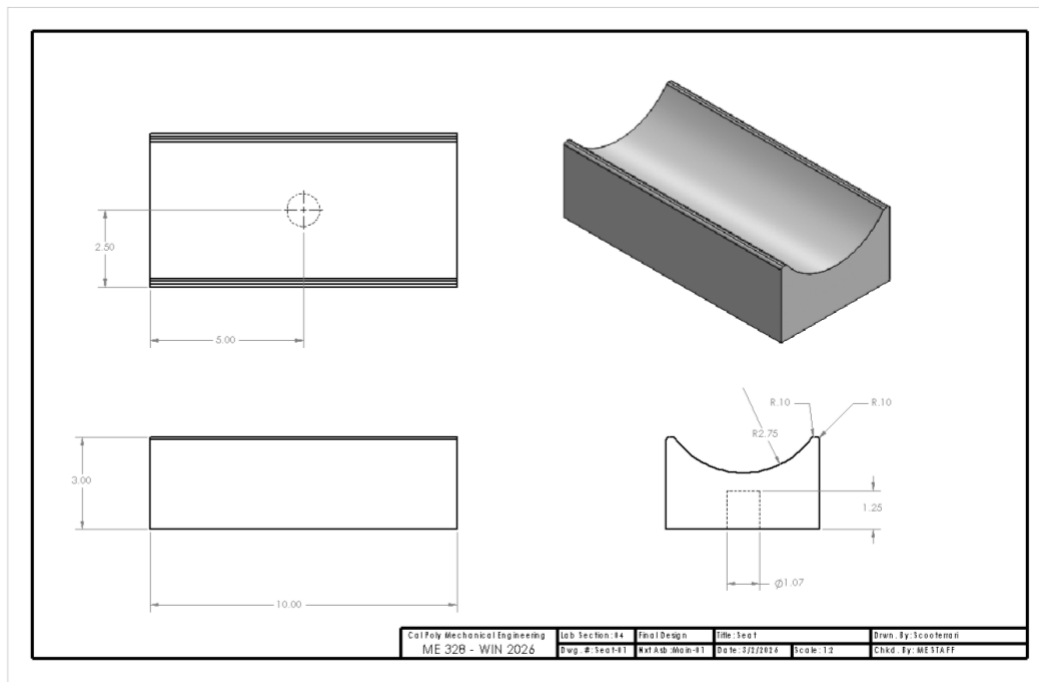


Figure 6. 3D Printed Seat Drawing.

Recommendations for Assembly Process:

Cut lengths should not be pre-measured sequentially on a single PVC member, as each cut removes material equal to the blade width and introduces cumulative dimensional error. This tolerance stack-up can result in undersized components and misalignment during assembly. Each segment should instead be measured individually after every cut to maintain dimensional accuracy.

Special care should be taken when drilling the 5/8" axle holes in the front and rear vertical PVC members. Given the steering column's 3/4" internal diameter, removing a 5/8" section significantly reduces remaining wall thickness and increases the risk of cracking. The hole should therefore be drilled incrementally using progressively larger bits to reduce fracture risk. Additionally, the rear vertical members must be carefully aligned prior to bonding to ensure proper axle fit and prevent binding.

The frame should be first assembled without any glue to ensure correct dimensions and fitment. Then the frame can be glued together. When doing so, it is good to bond one section at a time. For example, bonding such as two fittings and one tube together and then place it back on the full assembly, and make adjustments, so that each section gets bonded in the correct orientation. This reduces the chance that the frame becomes skewed, bent, or the fittings which have tubes running through them, for the steering and seat post, become not perpendicular to the frame.

Cost Breakdown:

Table 2. Itemized Cost of the Scooterrari Knee Scooter

Store	Items Purchased	Price (\$)
Home Depot	10' 1" PVC	7.31
	10' 3/4" PVC	4.86
	1" PVC Elbow Fitting x2	8.92
	1" PVC T Fitting x2	2.52
	1" PVC 90 Degree Fitting x2	3.21
	3/4" PVC T Fitting	0.73
	1' 5/8-11 Threaded Rod x2	9.54
	5/8-11 Nuts x4	1.88
	1/4" Pins x2	9
Harbor Freight	8" Wheels x3	22.47
Ace Hardware	1 Inch PVC Cross Fitting x2	2
	Foam	6
Total		77
After Reusable Item Return		26.58

Hand Calculations:

The selected factors of safety were determined based on load magnitude, structural function, and the consequences of failure. A higher factor of safety (3–4) was applied to the seat post and horizontal connecting members because these components carry the primary vertical load of the user and experience the largest bending moments. As the main load-bearing elements in the structural load path, failure of these members would result in loss of support and pose a significant injury risk. The elevated safety factor accounts for uncertainties in loading conditions, including dynamic effects during mounting, dismounting, and maneuvering.

In contrast, the steering column is subjected to comparatively lower loads and does not serve as the primary weight-supporting component. Therefore, a lower factor of safety in the range of approximately 1.2–2 was selected to maintain structural adequacy while reducing material usage and overall cost. This approach reflects a risk-based design strategy that prioritizes safety in critical structural members while optimizing efficiency in secondary components.

Table 3. PVC Pipe Dimensions and Sectional Properties

Sold as Size (in)	PVC OD (in)	PVC ID (in)	Area (A) (in ²)	Moment of Inertia (I) (in ⁴)	Distance to the Outermost Fiber, c (in)
0.75	1.05	0.75	0.424	0.044	0.525
1.0	1.315	1.0	0.572	0.098	0.6575
1.5	1.9	1.61	0.799	0.310	0.95

PVC Elastic Modulus = 349541 psi

PVC Yield Strength = 6000 psi

Seat post Calculations

- Maximum axial stress

$$\sigma = \frac{F}{A} \quad (1.1)$$



$$A = \frac{\pi}{4} (d_o^2 - d_i^2) \quad (1.2)$$

Axial stress calculations for the 0.75-inch diameter seat post PVC:

1. Normal Case

$$\sigma = \frac{160 \text{ lbf}}{0.424 \text{ in}^2} = 377 \text{ psi} \quad (1.3)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{377 \text{ psi}} = 15.9 \quad (1.4)$$

2. Abuse Case

$$\sigma = \frac{320 \text{ lbf}}{0.424 \text{ in}^2} = 755 \text{ psi} \quad (1.5)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{755 \text{ psi}} = 7.95 \quad (1.6)$$

Axial stress calculations for the 1.0-inch diameter seat post PVC:

1. Normal Case

$$\sigma = \frac{160 \text{ lbf}}{0.572 \text{ in}^2} = 279.5 \text{ psi} \quad (1.3)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{279.5 \text{ psi}} = 21.47 \quad (1.4)$$

2. Abuse Case

$$\sigma = \frac{320 \text{ lbf}}{0.572 \text{ in}^2} = 559 \text{ psi} \quad (1.5)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{559 \text{ psi}} = 10.7 \quad (1.6)$$

b. Critical Buckling load



Figure 7. Schematic of Fixed-Free Load Case.

$$P_{cr} = \frac{\pi^2 EI}{L_e^2} \quad (2.1)$$

Critical buckling load calculations for the 0.75-inch diameter seat post PVC:

$$P_{cr} = \frac{\pi^2 (349541 \text{ psi}) (0.044 \text{ in}^4)}{(2 * 8 \text{ in})^2} = 593 \text{ lbf} \quad (2.2)$$

Critical buckling load calculations for the 1.0-inch diameter seat post PVC:

$$P_{cr} = \frac{\pi^2 (349541 \text{ psi}) (0.098 \text{ in}^4)}{(2 * 8 \text{ in})^2} = 1321 \text{ lbf} \quad (2.3)$$

Horizontal Connecting Pipes Calculations

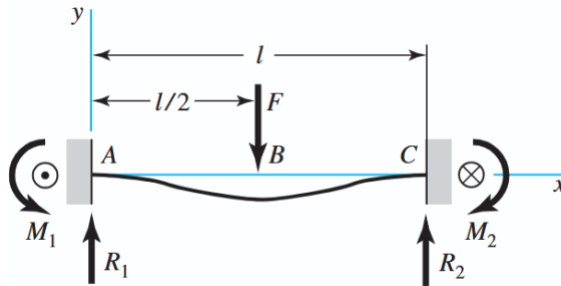


Figure 8. Schematic of Fixed-Fixed Load Case.

a. Maximum bending stress

$$\sigma = \frac{Mc}{I} \quad (3.1)$$

$$M = \frac{Fl}{8} \quad (3.2)$$

$$c = \frac{d_o}{2} \quad (3.3)$$

$$I = \frac{\pi}{64} (d_o^4 - d_i^4) \quad (3.4)$$

Note: The calculated stress is both the normal and principal stress. Transverse shear does not apply due to L/h being greater than 10. Therefore, according to the distortion energy theory,

$$\sigma' = \sqrt{\sigma_x^2} = \sigma_x \quad (3.5)$$

Maximum bending stress calculations for the 1.0-inch diameter horizontal PVC:

1. Normal Case

$$\sigma = \frac{Mc}{I} = \frac{240 \text{ in} * \text{lb} (.6575 \text{ in})}{0.098 \text{ in}^4} = \mathbf{1616 \text{ psi}} \quad (3.6)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{1616 \text{ psi}} = 3.71 \quad (3.7)$$

2. Abuse Case

$$\sigma = \frac{Mc}{I} = \frac{(480 \text{ in} * \text{lb} f)(.6575 \text{ in})}{0.098 \text{ in}^4} = 3232 \text{ psi} \quad (3.8)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{3220 \text{ psi}} = 1.86 \quad (3.9)$$



Maximum bending stress calculations for the 1.5-inch diameter horizontal PVC:

1. Normal Case

$$\sigma = \frac{Mc}{I} = \frac{(240 \text{ in} * \text{lbf})(.95 \text{ in})}{0.310 \text{ in}^4} = 736.1 \text{ psi} \quad (3.10)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{736.1 \text{ psi}} = 8.2 \quad (3.11)$$

2. Abuse Case

$$\sigma = \frac{Mc}{I} = \frac{(480 \text{ in} * \text{lbf})(.95 \text{ in})}{0.310 \text{ in}^4} = 1472.2 \text{ psi} \quad (3.12)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{1472.2 \text{ psi}} = 4.1 \quad (3.13)$$

Maximum deflection calculations for the 1.0-inch diameter horizontal PVC:

$$y_{max} = -\frac{Fl^3}{192EI} \quad (4.1)$$

1. Normal Case

$$y_{max} = -\frac{(80 \text{ lbf})(24 \text{ in})^3}{192(349541 \text{ psi})(0.098 \text{ in}^4)} = \mathbf{0.168 \text{ in}} \quad (4.2)$$

2. Abuse Case

$$y_{max} = -\frac{(160 \text{ lbf})(24 \text{ in})^3}{192(349541 \text{ psi})(0.098 \text{ in}^4)} = 0.337 \text{ in} \quad (4.3)$$

Maximum deflection calculations for the 1.5-inch diameter horizontal PVC:

1. Normal Case

$$y_{max} = -\frac{(80 \text{ lbf})(24 \text{ in})^3}{192(349541 \text{ psi})(0.310 \text{ in}^4)} = 0.053 \text{ in} \quad (4.4)$$



2. Abuse Case

$$y_{max} = -\frac{(160 \text{ lbf})(24 \text{ in})^3}{192(349541 \text{ psi})(0.310 \text{ in}^4)} = 0.106 \text{ in} \quad (4.5)$$

Steering Column Calculations

a. Axial Stress

Axial stress calculations for the 0.75-inch diameter steering column PVC:

1. Normal Case

$$\sigma = \frac{40 \text{ lbf}}{0.424 \text{ in}^2} = \mathbf{94.34 \text{ psi}} \quad (5.1)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{94.34 \text{ psi}} = 63.6 \quad (5.2)$$

2. Abuse Case

$$\sigma = \frac{80 \text{ lbf}}{0.424 \text{ in}^2} = 188.68 \text{ psi} \quad (5.3)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{188.68 \text{ psi}} = 31.8 \quad (5.4)$$

Axial stress calculations for the 1.0-inch diameter steering column PVC:

1. Normal Case

$$\sigma = \frac{40 \text{ lbf}}{0.572 \text{ in}^2} = 69.93 \text{ psi} \quad (5.5)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{69.93 \text{ psi}} = 85.8 \quad (5.6)$$

2. Abuse Case

$$\sigma = \frac{80 \text{ lbf}}{0.572 \text{ in}^2} = 139.86 \text{ psi} \quad (5.7)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{139.86 \text{ psi}} = 42.9 \quad (5.8)$$

b. Critical Buckling load



Figure 9. Schematic of Fixed-Free Buckling Load Case.

$$P_{cr} = \frac{\pi^2 EI}{L_e^2} \quad (6.1)$$

Critical buckling load calculations for the 0.75-inch diameter steering column PVC:

$$P_{cr} = \frac{\pi^2 (349541 \text{ psi})(0.044 \text{ in}^4)}{(2 * 28 \text{ in})^2} = \mathbf{48.40 \text{ lbf}} \quad (6.2)$$

Critical buckling load calculations for the 1.0-inch diameter steering column PVC:

$$P_{cr} = \frac{\pi^2 (349541 \text{ psi})(0.098 \text{ in}^4)}{(2 * 28 \text{ in})^2} = 107.81 \text{ lbf} \quad (6.3)$$

c. Maximum bending stress

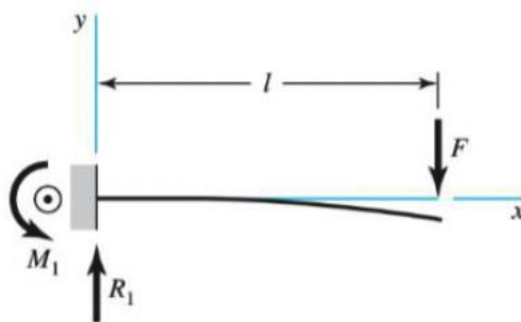


Figure 10. Schematic of Cantilever Load Case.



$$\sigma = \frac{Mc}{I} \quad (7.1)$$

$$M = Fl \quad (7.2)$$

$$c = \frac{d_o}{2} \quad (7.3)$$

$$I = \frac{\pi}{64} (d_o^4 - d_i^4) \quad (7.4)$$

Note: The calculated stress is both the normal and principal stress. Transverse shear does not apply due to L/h being greater than 10. Therefore, according to the distortion energy theory,

$$\sigma' = \sqrt{\sigma_x^2} = \sigma_x \quad (7.5)$$

Maximum bending stress calculations for the 0.75-inch diameter steering column PVC:

1. Normal Case

$$\sigma = \frac{Mc}{I} = \frac{(20 \text{ lbf})(25 \text{ in})(0.525 \text{ in})}{0.044 \text{ in}^4} = 5965.91 \text{ psi} \quad (7.6)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{5965.91 \text{ psi}} = 1.01 \quad (7.7)$$

2. Abuse Case

$$\sigma = \frac{Mc}{I} = \frac{(40 \text{ lbf})(25 \text{ in})(0.525 \text{ in})}{0.044 \text{ in}^4} = 11931.82 \text{ psi} \quad (7.8)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{11931.82 \text{ psi}} = 0.503 \quad (7.9)$$

Maximum bending stress calculations for the 1.0-inch diameter steering column PVC:

1. Normal Case

$$\sigma = \frac{Mc}{I} = \frac{(20 \text{ lbf})(25 \text{ in})(.6575 \text{ in})}{0.098 \text{ in}^4} = 3354.59 \text{ psi} \quad (7.10)$$



$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{3354.59 \text{ psi}} = 1.79 \quad (7.11)$$

2. Abuse Case

$$\sigma = \frac{Mc}{I} = \frac{(40 \text{ lbf})(25 \text{ in})(.6575 \text{ in})}{0.098 \text{ in}^4} = 6709.18 \text{ psi} \quad (7.12)$$

$$S.F. = \frac{S_y}{\sigma} = \frac{6000 \text{ psi}}{6709.18 \text{ psi}} = 0.894 \quad (7.13)$$

d. Maximum deflection

$$y_{max} = -\frac{Fl^3}{3EI} \quad (8.1)$$

Maximum deflection calculations for the 0.75-inch diameter steering column PVC:

1. Normal Case

$$y_{max} = -\frac{(20 \text{ lbf})(25 \text{ in})^3}{3(349541 \text{ psi})(0.044 \text{ in}^4)} = \mathbf{6.77 \text{ in}} \quad (8.2)$$

2. Abuse Case

$$y_{max} = -\frac{(40 \text{ lbf})(25 \text{ in})^3}{3(349541 \text{ psi})(0.044 \text{ in}^4)} = 13.55 \text{ in} \quad (8.3)$$

Maximum deflection calculations for the 1.0-inch diameter steering column PVC:

1. Normal Case

$$y_{max} = -\frac{(20 \text{ lbf})(24 \text{ in})^3}{3(349541 \text{ psi})(0.097 \text{ in}^4)} = 3.04 \text{ in} \quad (8.4)$$

2. Abuse Case

$$y_{max} = -\frac{(40 \text{ lbf})(24 \text{ in})^3}{3(349541 \text{ psi})(0.097 \text{ in}^4)} = 6.08 \text{ in} \quad (8.5)$$

FEA Modeling and Simulation Results:

Horizontal Column FEA

The structural integrity of the horizontal beam was evaluated through a dual-verification approach, comparing hand calculations to a simulated Finite Element Analysis (FEA) model. The analytical hand calculations predicted a maximum stress of 1,616 psi and a peak deflection of 0.17 in, establishing a baseline for the beam's expected behavior under loading. These values were validated by the FEA results, which yielded a nearly identical maximum stress of 1,615 psi and a peak deflection of 0.18 in. The high degree of correlation between these two methods, demonstrating a variance of less than 0.1% for stress, confirms the accuracy of the structural model and provides a high level of confidence in the beam's design and safety margins.

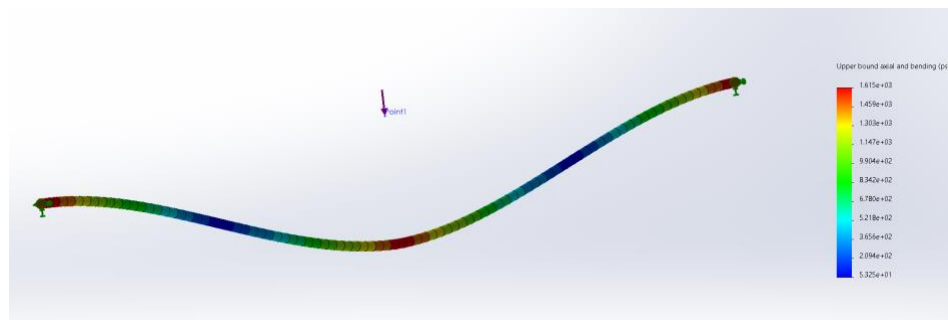


Figure 11. Normal Stress and Bending Moments for Horizontal Beam (Beam Element Model).

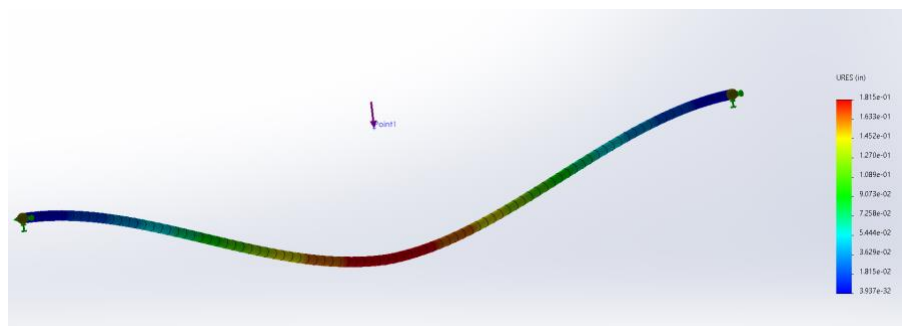


Figure 12. Total Displacement Heat Map for Horizontal Beam (Beam Element Model).

Note: FEA revealed that initial hand calculations relied on incorrect assumptions, which had resulted in an over-engineered design. By correcting these models, the beam was successfully optimized for a smaller 1-in. vs 1.5-in PVC tube, with a lighter profile without compromising structural integrity.

Steering Column FEA

The structural integrity of the steering column assembly was evaluated using a beam element model to simulate critical loading scenarios. Through Finite Element Analysis (FEA), the design's resistance to elastic instability was quantified, yielding a buckling factor of safety of 4.77. This margin indicates the column's proximity to the theoretical buckling limit under axial loads. Furthermore, the bending performance was assessed to ensure the assembly maintains sufficient stiffness and strength; the peak bending stress was recorded at 3.3 ksi, while the maximum resulting bending displacement reached 3.6 inches. These results, illustrated in Figures 5 through 7, provide the data for validating the column's current geometry against performance requirements.

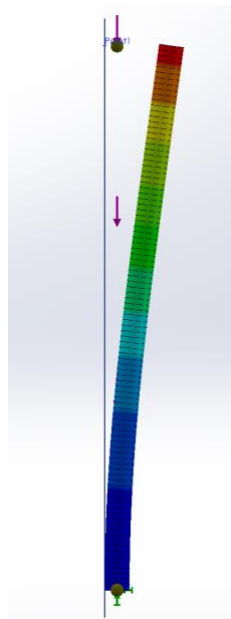


Figure 13. Buckling FEA for Steering Column (Beam Element Model).

Buckling factor of safety: 4.7744

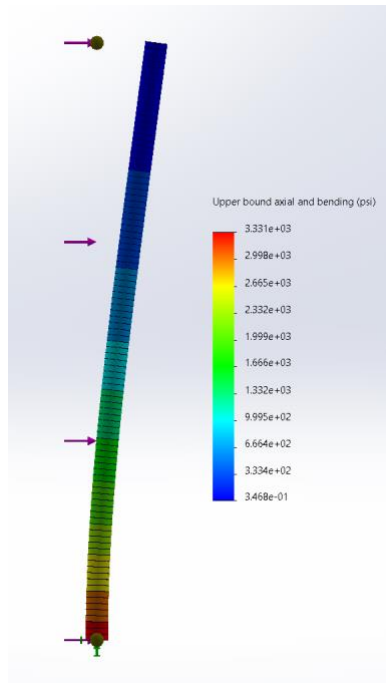


Figure 14. Bending Stress Map for Steering Column (Beam Element Model).

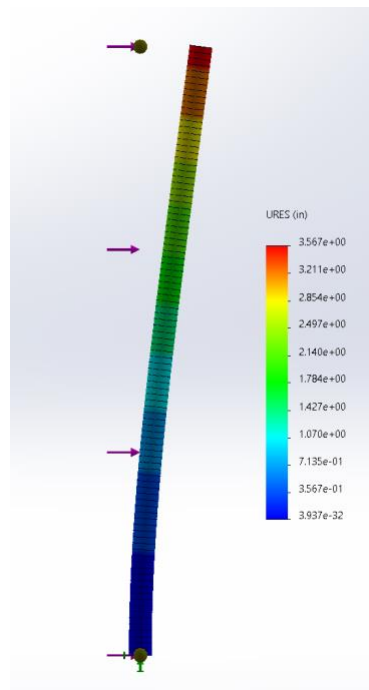


Figure 15. Bending Displacement Map for Steering Column (Beam Element Model).

Seat Post FEA

A finite element analysis (FEA) was conducted on the seat post using a linear buckling study in SolidWorks to evaluate structural stability under compressive loading. The analysis produced a buckling factor of safety of 3.3859, indicating that the predicted critical buckling load is approximately 3.39 times greater than the applied load of 160 lbf. This result, shown in figure 8, demonstrates a significant margin against elastic buckling under the specified loading conditions and supports the structural adequacy of the seat post design.

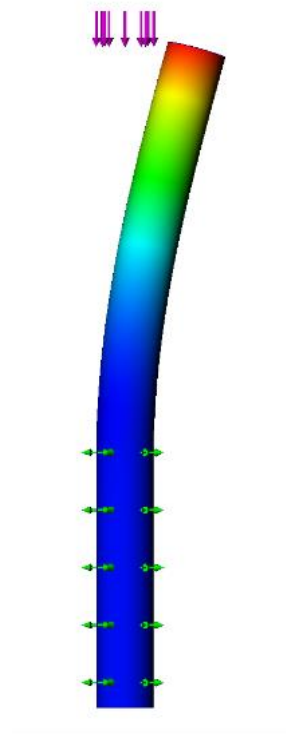


Figure 16. Buckling FEA for Seat Column.

Buckling factor of safety: 3.3859

Critical Points: Deflection and Failure Modes

Buckling Stability: The final analysis yields a Buckling Factor of Safety of 4.77 for the steering column and 3.38 for the seat post. These values significantly exceed the threshold for failure, ensuring that the assembly will not collapse under peak axial loading.

Under extreme bending scenarios, the steering column showed a maximum displacement of 3.6 inches at a peak stress of 3.33 ksi. While this displacement appears high, and we plan on taking



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steps to reduce this, it represents the theoretical limit under peak loading; in standard operation, the stiffness should be sufficient to maintain stability.

One of the most important and critical points is at the midpoint of the scooter, that is where the largest bending moment takes place, and therefore the most stress.

Deflection Measurements and Other Testing Results:



Figure 17. Initial no load deflection on the horizontal beam.



Figure 18. 160lb deflection on the horizontal beam.

As shown in figures 15 and 16, the deflection on the horizontal beam was a half inch. While this is greater than we wanted it to be, we believe another contributor to this number was the wheels pushing outward from the frame, resulting in a larger displacement.

The FEA resulted in a displacement of only .182 inches, while our observations resulted in a half inch. We believe this to be because of differences in boundary conditions. The scooter being modeled as fixed-fixed is an idealized situation, realistically the scooter has more complex loading occurring.



Figure 19. Eamon riding the scooter

As is visible from the picture, the dimensions were optimal for Eamon, who the scooter was designed for, including the height of the handlebars and seat post, and the distance from the steering column to the seat post. The total length of the scooter did work well but could be a couple inches shorter to decrease mass. The knee pad was very comfortable with foam attached.

Recommendations for Improvement:

Future design iterations should focus on increasing structural stiffness, reducing deflection, and improving overall stability and steering performance. Increasing the chassis tubing diameter from 1 inch to 1.5 inches would substantially increase the area moment of inertia, resulting in a significant reduction in bending stress and vertical deflection under rider loading. Additionally, shortening the horizontal frame members by approximately 3–6 inches would reduce the effective moment arm, thereby decreasing peak bending moments and improving global frame rigidity. Steering performance could be enhanced by incorporating rolling-element bearings within the



steering column and wheel axles to reduce frictional losses, improve rotational smoothness, and enhance directional stability, thereby mitigating tipping risk during maneuvering. Replacing the current horizontal “side-by-side” bar configuration with a vertically stacked (over-under) arrangement would better align structural members with the primary loading axis, increasing bending stiffness in the dominant load direction. Implementing a direct structural connection between the seat post and steering column would further create a more continuous load path, reducing compliance within the frame and improving torsional rigidity and rider control. Also, transitioning from a three-wheel to a four-wheel configuration would increase the base of support and improve static and dynamic stability, particularly during turning maneuvers where lateral load transfer can induce tipping in narrower three-wheel geometries. Finally, having a tee fitting on the bottom of the steering column with spacers to hold the axle would provide more stability than putting the axle through a hole in the bottom of the column.

Conclusion and Reflection:

The Scooterrari project successfully met its primary objective of designing a functional, low-cost knee scooter with a user capacity. However, the transition from theoretical modeling to physical testing revealed several gaps, most notably a horizontal beam deflection that exceeded our initial design limit of 3.6 inches. This large steering column displacement, along with the difficulty of three wheels, led to an unstable ride. The project was a good learning experience, and reflecting on these results, it is clear that while our FEA models were mathematically accurate, they did not fully account for real-world assembly variables that also affected the performance of the scooter.

Despite these challenges, we believe the Scooterrari remains a viable concept. With strategic improvements, such as transitioning to a four-wheel configuration, increasing tubing diameter to 1.5 inches, and incorporating rolling-element bearings, the design could achieve a better ride with more rigid stability like we had originally envisioned. This project served as a valuable lesson in the critical balance between material selection and structural rigidity, providing a clear engineering roadmap for future iterations of the platform.